

Fisher 500-C Stereophonic FM Multiplex Receiver — Project Reference

Manufacturer: Fisher Radio Corporation

Production: 1963–1964

Original MSRP: \$369 ($\approx$ \$2,325 in 2005 dollars)

Estimated Production: 100,000+ units

Overview

The Fisher 500-C is a vacuum tube stereo FM multiplex receiver designed by Avery Fisher. It combines a full-function preamplifier, a 30 watt-per-channel power amplifier, and a high-sensitivity stereo FM tuner in a single chassis. It was the flagship of Fisher's receiver line in 1964, and remains one of the most sought-after vintage tube receivers among collectors and audiophiles. The 500-C was a direct evolution of the 500-B, with the primary upgrade being the addition of a front-panel headphone jack and a numbered signal-strength indicator (replacing the 500-B's "Stereo Beam" magic-eye tube).

Specifications

Amplifier Section

Parameter	Value
Type	Stereo FM Receiver (tube)
Power Output (rated)	30 W/ch into 8 Ω (stereo), 0.7% THD
Power Output (music power)	75 W/ch total music power
Power at Clipping (1% THD+N)	36 W into 8 Ω · 26 W into 16 Ω · 6 W into 4 Ω
Frequency Response	25 Hz – 25 kHz, $\pm$ 1.5 dB
IM Distortion (60 Hz/7 kHz)	0.7% (4:1 ratio)
Signal-to-Noise Ratio	80 dB (line input)
Channel Separation	50 dB (line input)

Parameter	Value
Voltage Gain (max, Aux)	38.4 dB into 8 Ω at 1 kHz
Output Impedance	~2 Ω (bass/midrange), rising to 9.3 Ω at 20 kHz
Input Impedance	Spec: 100 k Ω · Measured: 180 k Ω (20 Hz), 470 k Ω + (1 kHz), 300 k Ω (20 kHz)
Output Transformer Taps	4 Ω , 8 Ω , 16 Ω
Rectification	Solid-state full-wave (front end), half-wave (B+ rails), RC Pi filtering

FM Tuner Section

Parameter	Value
Tuning Range	FM only (no AM; see 800-C for AM)
IHF FM Usable Sensitivity	1.8 μ V
Signal-to-Noise Ratio	70 dB
Selectivity	60 dB
FM Channel Separation	35 dB
Multiplex Decoder	Fisher standard MPX (shared across Fisher tuner line)
FM Front End	Fisher "Golden Synchrode" — RF amplifier tube + 3 tuned circuits

Input Sensitivities

Input	Sensitivity
Phono (MM)	3.3 mV
DIN	450 mV
Line (Aux)	230 mV

Physical

Parameter	Value
Dimensions	17.5" W × 6" H × 13" D (without case)
Dimensions (alt)	18.5" W × 7.25" H × 15" D (with case)
Weight	38 lbs (without wood case)
Power Requirement	105-120 V AC, 50/60 Hz
Power Consumption	235 W (full output), 270 VA
Optional Cabinet	Walnut or Mahogany wood — \$24.95 in 1964

Tube Complement

Tube Type	Quantity	Function
12AX7	7	Phono preamp, tone control, phase inverter stages
7591	4	Output stage (push-pull, 2 per channel)
6HA5	1	Tuner section
6HR6	1	Tuner section
6CW4	2	Tuner section (nuvistor)
6AU6 / EF94	3	Tuner section

Total: 18 tubes

Notes on tube complement:

- The chassis has 9 visible positions for small tubes which can cause confusion vs. the spec listing $7 \times 12AX7$ — the extra sockets are for the tuner section tubes.
- The phono/tape-head stage uses one 12AX7 per channel for approximately 20 dB of amplification.
- Current production Tung-Sol 7591A output tubes are widely regarded as the closest to original specs and run more efficiently than Electro-Harmonix alternatives.

- JJ-branded 7591 replacements have drawn reliability concerns from the restoration community.
 - Original Fisher-branded tubes were manufactured by various suppliers (RCA, Telefunken, Tung-Sol, etc.).
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Inputs, Outputs & Controls

Rear Panel Inputs (6 pairs of stereo inputs)

1. **Phono High** — For ceramic cartridges
2. **Phono Low** — For mid- to high-output magnetic (MM) cartridges
3. **Tape Head** — Direct tape head input (uses phono stage with different EQ)
4. **Monitor** — High-level input for tape monitor playback
5. **Aux** — High-level auxiliary input
6. **Reverberation (Spacexpander)** — Connections for Fisher K-10 Spacexpander; includes jumper plugs on chassis top. **Critical: If jumpers are missing, the receiver will not produce sound.**

Rear Panel Outputs

- **Tape Record Output** — Dedicated recording output pair
- **Speaker A** — First set of speaker terminals (screw-type)
- **Speaker B** — Second set of speaker terminals (screw-type)
- **Center Channel** — Connection for center-channel speaker
- **Output Transformer Taps:** 4 Ω , 8 Ω , 16 Ω (per channel)

Front Panel Controls

- **Tuning Knob** (upper left) — FM dial
- **Signal Strength Indicator** — Numbered meter (replaced 500-B's magic-eye "Stereo Beam" tube)
- **Stereo Indicator** — Relay-powered light for FM stereo broadcasts
- **Volume / Power** — Combined on/off switch and volume (note: AC/volume switches are known to fail)
- **Bass** — Tone control
- **Treble** — Tone control

- **Loudness Contour** — Switch for Fletcher-Munson compensation
 - **Balance** — Channel balance
 - **Mode Switch** — AM / FM / FM Stereo / Phono / Tape / Aux
 - **Speaker Selector** — Speaker 1, Speaker 2, or Speaker 1+2
 - **Headphone Jack** — Center-mounted, $\frac{1}{4}$ " — designed for direct monitoring during tape recording
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Measured Performance Notes (Stereophile, John Atkinson)

These measurements were performed on a representative restored unit (S/N 58022S) with EH 7591A output tubes, silicon bridge rectifier, and bias set to 34–35 mA:

- **Polarity:** Non-inverting
 - **Channel Balance:** Excellent at 12:00 volume; degrades at extremes (>1 dB imbalance at high/low settings)
 - **Bass Response:** Rolls off sharply below 35 Hz even with high-pass filter bypassed
 - **Treble Response:** Down 5 dB at 20 kHz into $8\ \Omega$ — significant top-octave energy loss
 - **Output Impedance Effect:** ± 1.6 dB response variation into simulated speaker loads (audible)
 - **$4\ \Omega$ Operation:** Very limited — output tube plates glow cherry red above ~ 1 W; only 6 W at clipping. Best paired with $8\ \Omega$ or higher impedance speakers.
 - **Squarewave:** 10 kHz shows long risetimes and rounded shape; 1 kHz shows small overshoot (well-damped parasitic resonance)
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Common Issues & Restoration Notes

Known Failure Points

1. **Selenium Rectifier** — The original selenium bridge rectifier is failure-prone and should be replaced with a silicon bridge rectifier (e.g., RadioShack #276-1152). Early serial numbers have selenium; later production units (70000+ S/N) already have silicon.
2. **Electrolytic Capacitors** — All original electrolytics should be replaced, especially the large dual 1000 μ F bias-filter capacitor (known hum source).
3. **AC/Volume Switch** — Known to fail; many owners use a power strip for on/off to avoid the switch.

4. **Coupling Capacitors** — Replace with 0.1 μF / 600 V polypropylene film caps.
5. **Knob Brass Caps** — Frequently fall off and get lost. Replacements are available. All original caps have an indicator line except the tuning knob.
6. **Line Voltage** — Original design was for 105–120 V; modern wall voltage (closer to 120–125 V) can stress components. A variac or bucking transformer may be advisable.
7. **Spacexpander Jumpers** — Must be present on the top-chassis connectors or the unit will produce no sound.

Recommended Restoration Modifications

- **Replace selenium rectifier** with silicon bridge + series resistor to compensate for voltage increase
- **Install cathode resistors** (10 Ω , 0.5 W) on each output tube socket — protects output transformers and allows easy bias measurement
- **Adjust bias** by changing shunt resistor R131 — target 34–35 mA per output tube (good compromise for EH 7591A reissues)
- **Add 220 k Ω grid resistors** to output tube bias board
- **Replace all electrolytic and film capacitors** in: bias supply, power supply, amplifier section, FM signal path, and multiplex section
- **Consider high-voltage delay** (SCR-based) to extend tube life and reduce harmful inrush current
- **Clean all switches and pots** with DeoxIT and Cali-Lube (Caig Chemicals)
- **Align FM tuner and stereo MPX** after recapping the tuner section
- **Optional:** Reduce DC filament voltage on phono-stage 12AX7 tubes by adding 5 Ω / 5 W resistor in series with R142 (changes effective value from 15 to 20 Ω)

Restoration Kit Sources

- **Metalbone (metalbone.net)** — Full restoration kits with Orange Drop coupling caps, installation guide with photos, reverse tube layout diagram, schematic copies. ~\$85 + \$11 shipping.
- **FisherDoctor (Al Pugliese)** — RESTO-PAKs with caps, resistors, and all necessary parts. fisherdoc@aol.com / (718) 948-7489
- **AEA Audio (aeaaudio.com)** — High-end full restoration service; custom modifications for audiophile-grade performance
- **VTV (Vintage Tube Vendor)** — Restoration kits available (community opinions on value vary)

- **Hudson Valley HiFi** — Documented professional restorations with detailed photo blogs

Related Fisher Models

Model	Relationship to 500-C	Price (1964)
Fisher 500-B	Direct predecessor; magic-eye tuning tube; no headphone jack	—
Fisher 400	FM stereo receiver; 65 W total; budget alternative	\$309
Fisher 800-C	Identical to 500-C but adds AM tuner	\$429
Fisher 600	Transistorized FM receiver; 100 W output	\$595

Market Value (approximate, as of research date)

Condition	Price Range
Unrestored	~\$700
Fully restored, good cosmetics	~\$1,500 – \$2,200

Key Reference Links

Resource	URL
Stereophile Full Review	https://www.stereophile.com/historical/605fisher/index.html
Stereophile Specifications	https://www.stereophile.com/content/fisher-500-c-vintage-stereo-receiver-specifications
Stereophile Measurements	https://www.stereophile.com/content/fisher-500-c-vintage-stereo-receiver-measurements
Classic Receivers Overview	https://classicreceivers.com/fisher-500c

Resource	URL
HiFi Engine Manual Library	https://www.hifiengine.com/manual_library/fisher/500-c.shtml
Owner's Manual (PDF)	https://www.thetubestore.com/lib/thetubestore/schematics/Fisher/Fisher-500C-Owners-Manual.pdf
Schematic (PDF)	https://www.thetubestore.com/lib/thetubestore/schematics/Fisher/Fisher-500C-Schematic.pdf
Fisher Schematics Collection	https://www.thetubestore.com/fisher-schematics
Metalbone Restoration Kits	http://metalbone.net/fisher.html
AEA Signature 500-C Service	https://aeaaudio.com/fisher500c/
Hudson Valley HiFi Restoration	https://hudsonvalleyhifi.com/archives/948

Safety Warning

The Fisher 500-C uses potentially **lethal high voltages**. The B+ supply rails and output transformer circuits carry dangerous DC voltages. Any restoration, troubleshooting, or testing should only be performed by someone thoroughly familiar with safe practices around high-voltage tube equipment. Always discharge capacitors before working inside the chassis and never work alone.

Document compiled from Stereophile reviews, HiFi Engine, Classic Receivers, community restoration forums, and vendor documentation.